The empirical recurrence for OEIS A253039, recovered from its tail

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Abstract

OEIS A253039 records "Empirical recurrence of order 78" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 2592$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 81$. Its last coefficient is $c_{78} = 512 \neq 0$, so no further tail satisfies anything shorter; any order-78 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A253039 is "Number of (n+2) X (3+2) 0..3 arrays with every consecutive three elements in every row, column, diagonal and antidiagonal having exactly two distinct values, and new values 0 upwards introduced in row major order.". It has offset 1 and begins

$$928, 998, 1750, 3119, 6086, 12641, 25184, 54251, \ \ldots$$

The entry states:

> Empirical recurrence of order 78 (see link above).

As of the "Last modified" line on the live entry (Dec 15 2023 20:24:04, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 2592$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 2592$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 78: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 78.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 5184$ exact terms from $n = 3$ on, it returns order 78 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{78} c_i \, a(n-i),$$

c_1	0	c_{21}	2239	c_{41}	3473847	c_{61}	−282080
c_2	10	c_{22}	289610	c_{42}	−3353270	c_{62}	137240
c_3	7	c_{23}	587025	c_{43}	−2357834	c_{63}	39952
c_4	−7	c_{24}	219811	c_{44}	−5358300	c_{64}	400696
c_5	−70	c_{25}	59638	c_{45}	−3523330	c_{65}	79584
c_6	−234	c_{26}	−730807	c_{46}	2336602	c_{66}	−61280
c_7	65	c_{27}	−1266636	c_{47}	1411786	c_{67}	−13040
c_8	717	c_{28}	−545422	c_{48}	5141732	c_{68}	−103792
c_9	1394	c_{29}	−571620	c_{49}	2778692	c_{69}	−17024
c_{10}	1407	c_{30}	1519865	c_{50}	−1269404	c_{70}	20992
c_{11}	−3831	c_{31}	2331486	c_{51}	−712180	c_{71}	3680
c_{12}	−8882	c_{32}	1509238	c_{52}	−3927484	c_{72}	17600
c_{13}	−9589	c_{33}	1521195	c_{53}	−1692048	c_{73}	2688
c_{14}	2529	c_{34}	−2773302	c_{54}	593628	c_{74}	−4736
c_{15}	42380	c_{35}	−3185246	c_{55}	316100	c_{75}	−896
c_{16}	47263	c_{36}	−2891719	c_{56}	2376256	c_{76}	−1536
c_{17}	21349	c_{37}	−2617365	c_{57}	790896	c_{77}	0
c_{18}	−61146	c_{38}	3570497	c_{58}	−282676	c_{78}	512
c_{19}	−208761	c_{39}	3122003	c_{59}	−124456		
c_{20}	−129448	c_{40}	4433857	c_{60}	−1116616		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 81$, and it is the recurrence OEIS A253039 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{78} - c_1 x^{78-1} - \cdots - c_{78}$. Its constant term is $-c_{78} = -512$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 78 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 78, so $\deg q = 78$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 78, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 78, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 512.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-78 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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